

they over control. Unit-specific trends also increase the weight on units treated at the extremes of the panel, changing estimates for this reason as well.

The unilateral divorce analysis provides a striking illustration of how unit-specific trends can fail because figure 5 shows no pre-trends but strongly time-varying treatment effects. Trends shrink the estimate so much that its sign changes (0.59, s.e.=1.35). Table 2 shows that changes in 2x2 DD's account for 90 percent of the difference between estimates with and without trends, and panel A of figure 10 shows that most of the treated/untreated components are much less negative with trends, with especially large differences for the terms involving the 1970 states. Panel B of figure 10 shows that trends reduce the weight on $\hat{\beta}_{kU}^{2x2}$ terms, and increase the importance of timing-only comparisons. Since equation (11c) shows that the timing-only terms are already biased by the time-varying treatment effects, the change in weighting induced by unit-specific trends exacerbates this bias, accounting for 47 percent of the coefficient difference. Because unit-specific trends change the treated/untreated terms *and* give them less weight, the interaction of those two factors accounts for -36 percent of the overall change in the estimates.

v. *Group-specific linear pre-trends*

A simple strategy to address counterfactual trends is to estimate *pre-treatment* trends in Y_k^0 directly and partial them out of the full panel (cf. Bhuller et al. 2013, Goodman-Bacon 2018). This is what we hope that unit-specific trends do, it does not depend on the treatment effect pattern, and it does not change the weighting of the 2x2 DD's. Specifically, using data from before t_1^* , one can estimate a pre-trend in y_{it} for each timing group.³⁹ The slope will equal the linear component of

³⁹ One could estimate pre-trends for each *unit*, but this yields identical point estimates to the group-specific pre-trends. Moreover, group-specific trends reduce the variability of the estimator because trend deviations that would bias *unit* specific pre-trends cancel out to some extent when averaged by timing group. This matters because this strategy extrapolates potentially many periods into the future, magnifying specification error in the pre-trends. This point applies to unit-specific linear trend specifications as well. Point estimates are identical when this specification includes linear trends for each group rather than each unit.

unobservables plus a linear approximation to trend deviations before before t_1^* . Removing this trend from the full panel yields an outcome variable that is robust to linear trends, unaffected by time-varying treatment effects, and weights each 2x2 DD component in the same way as the unadjusted estimator. Like the unit-specific time trend control strategy, group-specific pre-trends are sensitive to non-linear unobservables in the pre-period. Appendix C analyzes this estimator in detail. Partialling out pre-trends yields a unilateral divorce effect of -6.52 (s.e. = 1.7); close to the average post-treatment effect of -4.92. While inference is outside the scope of this paper (see Athey and Imbens 2018), note that this two-step strategy necessarily involves a partly estimated outcome variable so second-stage standard errors are incorrect.

vi. *Disaggregated time effects*

If counterfactual outcomes evolve differently by a category, R , to which units belong, one can model these changes flexibly by including separate time fixed effects for each category:

$$y_{it} = \alpha_i + \alpha_t^{R(i)} + \beta_{R \times t}^{DD} D_{it} + e_{it} \quad (21)$$

The coefficient $\hat{\beta}_{R \times t}^{DD}$ equals an average of two-way fixed effects estimates by values of R weighted by the share of units in each R (n^R) and the within- R variance of $\tilde{D}_{it}$. For simplicity, I refer to R as “region”, but the analysis is not specific to region-by-time fixed effects. When treatments vary by county or city, for example, studies include state-by-time fixed effects (e.g. Almond, Hoynes, and Schanzenbach 2011, Bailey and Goodman-Bacon 2015); when treatments vary by firm studies include industry-by-year fixed effects (e.g. Kovak, Oldenski, and Sly 2018).

Since the decomposition theorem holds for each within-region DD estimate, it also holds for $\hat{\beta}_{R \times t}^{DD}$. Each 2x2 DD averages the region-specific 2x2 DD’s ($\hat{\beta}_{kU,R}^{2 \times 2}$ and $\hat{\beta}_{k\ell,R}^{2 \times 2}$) but the weights reflect region size and the within-region distribution of timing groups:

$$\hat{\beta}_{kU,R \times t}^{2 \times 2} = \sum_R \frac{n^R n_k^R n_U^R}{\sum_R n^R n_k^R n_U^R} \hat{\beta}_{kU,R}^{2 \times 2} \quad (22)$$

$$\hat{\beta}_{k\ell,R \times t}^{2 \times 2} = \sum_R \frac{n^R n_k^R n_\ell^R}{\sum_R n^R n_k^R n_\ell^R} [\mu_{k\ell} \hat{\beta}_{k\ell,R \times t}^{2 \times 2,k} + (1 - \mu_{k\ell}) \hat{\beta}_{k\ell,R \times t}^{2 \times 2,\ell}] \quad (23)$$

2x2 DD's from large regions get more weight (via the n^R), but the distribution of timing groups within region (the $n_a^R n_b^R$ terms) also determine the importance of each region. Regions with no units in group k contribute nothing to 2x2 DD's involving that group, in contrast to the simpler estimator where that region would contribute controls for group k . If no region contains units from a given pair of groups that pair drops out of the disaggregated time effects specification. These within-region 2x2 DD's in (22) and (23) are weighted together using the cross-region average of the sample share products, $\sum_R n^R n_k^R n_\ell^R$ as well as treatment variances.

Adding region-by-year effects to the unilateral divorce analysis cuts the estimated effect by a factor of three (-1.16). Figure 11 plots the 2x2 DD's and the weights from this specification against those from the baseline model. 43 of the 156 2x2 terms in the baseline model drop out of the within-region specification, and table 2 shows that about three quarters of the difference in the estimates comes from the way these fixed effects change the weights.

IV. CONCLUSION

Difference-in-differences is perhaps the most widely applicable quasi-experimental research design. Its transparency makes it simple to describe, test, interpret, and teach. This paper extends all of these advantages from canonical two-by-two DD models to general and much more common DD models with variation in the timing of treatment.

My central result, the DD decomposition theorem, shows that a two-way fixed effects DD coefficient equals a weighted average of all possible simple 2x2 DD's that compare one group that changes treatment status to another group that does not. Many ways in which the theoretical

interpretation of regression DD differs from the canonical model stem from the fact that these simple components are weighted together based both on sample sizes *and* the variance of their treatment dummy. This defines the DD estimand, the variance-weighted average treatment effect on the treated (VWATT), and generalizes the identifying assumption on counterfactual outcomes to variance-weighted common trends (VWCT). Moreover, I show that because already-treated units act as controls in some 2x2 DD's, the two-way fixed effects model requires an additional identifying assumption of time-invariant treatment effects.

The DD decomposition theorem also leads to several new tools for practitioners. Graphing the 2x2 DD's against their weight displays all the identifying variation in any DD application, and summing weights across types of comparisons quantifies “how much” of a given estimate comes from different sources of variation. I use the DD decomposition theorem to propose a reweighted balance test that reflects this identifying variation, is easy to implement, has higher power than tests of joint balance across groups, and shows how large and in what direction any imbalance occurs. I suggest several simple methods to learn why estimates differ across alternative specifications. The weighted average representation leads to a Oaxaca-Blinder-Kitagawa-style decomposition that quantifies how much of the difference in estimates comes from changes in the 2x2 DD's, the weights, or both. Plots of the components or the weights across specifications show clearly where differences come from and can help researchers understand why their estimates changes and whether or not it is a problem.